

fn make_np_sum

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This proof relies on an idealized model of numeric computation that ignores finite-machine limitations such as non-closure and overflow, so the corresponding API belongs behind the “**idealized-numeric**” feature flag.

This proof resides in “**contrib**” because it has not completed the vetting process.

Proves soundness of `make_np_sum` in `__init__.py` at commit `f5bb719` (outdated¹).

`make_np_sum` accepts an `input_domain` and `input_metric`, and returns a stable Transformation that computes the vector-valued sum with bounded L_p sensitivity.

1 Hoare Triple

Preconditions

None

Pseudocode

```
1
2 def make_np_sum(input_domain: Domain, input_metric: Metric) -> Transformation:
3     """Construct a Transformation that computes a sum over the row axis of a 2-dimensional
4     array.
5
6     :param input_domain: instance of 'array2_domain(size=_, num_columns=_)'
7     :param input_metric: instance of 'symmetric_distance()'
8
9     :return: a Measurement that computes the DP sum
10    """
11    import opendp.prelude as dp
12    np = import_optional_dependency('numpy')
13
14    dp.assert_features("contrib", "idealized-numeric")
15
16    if not str(input_domain).startswith("NumpyArray2Domain"): #
17        raise ValueError(f"input_domain ({input_domain}) must be NumpyArray2Domain") # pragma:
18        no cover
19
20    if input_domain.descriptor.nan:
21        raise ValueError(f"input_domain ({input_domain}) must not permit NaN elements") #
22        pragma: no cover
```

¹See new changes with `git diff f5bb719..32e0167 python/src/opendp/extras/numpy/_make_np_sum/__init__.py`

```

21     if input_metric != dp.symmetric_distance(): #
22         raise ValueError("input_metric must be the symmetric distance") # pragma: no cover
23
24     input_desc = input_domain.descriptor
25     norm = input_desc.norm
26     if norm is None: #
27         raise ValueError(f"input_domain ({input_domain}) must have bounds. See make_np_clamp")
28         # pragma: no cover
29
29     output_metric = {1: dp.l1_distance, 2: dp.l2_distance}[input_desc.p] #
30
31     if input_desc.size is None:
32         origin = np.atleast_1d(input_desc.origin)
33         norm += np.linalg.norm(origin, ord=input_desc.p)
34         stability = lambda d_in: d_in * norm
35     else:
36         stability = lambda d_in: d_in // 2 * 2 * norm
37
38     return _make_transformation(
39         input_domain,
40         input_metric,
41         dp.vector_domain(dp.atom_domain(T=input_desc.T, nan=False)),
42         output_metric(T=input_desc.T),
43         lambda arg: arg.sum(axis=0), #
44         stability,
45     )

```

Postcondition

Theorem 1.1. For every setting of the input parameters (`input_domain`, `input_metric`) to `make_np_sum` such that the given preconditions hold, `make_np_sum` raises an error (at compile time or run time) or returns a valid transformation. A valid transformation has the following properties:

1. (Data-independent runtime errors). For every pair of members x and x' in `input_domain`, `invoke(x)` and `invoke(x')` either both return the same error or neither return an error.
2. (Appropriate output domain). For every member x in `input_domain`, `function(x)` is in `output_domain` or raises a data-independent runtime error.
3. (Stability guarantee). For every pair of members x and x' in `input_domain` and for every pair (d_{in}, d_{out}) , where d_{in} has the associated type for `input_metric` and d_{out} has the associated type for `output_metric`, if x, x' are d_{in} -close under `input_metric`, `stability_map(d_in)` does not raise an error, and `stability_map(d_in) = d_out`, then `function(x), function(x')` are d_{out} -close under `output_metric`.

Proof. Let x and x' be arbitrary members of `input_domain`. By 15, `input_domain` is an `NArray2Domain`, and by 21, `input_metric` is `SymmetricDistance`. Let R denote `input_domain.descriptor.norm`, and let O denote `input_domain.descriptor.origin`. By 26, R is defined, and by 29, the output metric is the L^p metric with $p \in \{1, 2\}$. Because this proof is under idealized-numeric, we ignore machine arithmetic issues such as overflow and treat vector addition as associative and commutative.

Data-independent runtime errors. All runtime checks in `make_np_sum` occur before the transformation is returned. For any member of `input_domain`, the function on line 43 applies `sum(axis=0)` to a 2-dimensional numpy array, which is defined for every such input. Therefore, for every pair $x, x' \in \text{input_domain}$, neither `invoke(x)` nor `invoke(x')` raises a runtime error.

Appropriate output domain. Each member of `input_domain` is a 2-dimensional numpy array whose entries have type `input_domain.descriptor.T` and do not contain NaNs. Applying `sum(axis=0)` removes

axis zero and returns a 1-dimensional numpy array of the same atomic type. Under idealized numerics, summing non-NaN values yields non-NaN values. Hence `function(x)` is a member of `output_domain`.

Stability guarantee. Fix any `d_in` and `d_out` such that x and x' are `d_in`-close under `input_metric` and `stability_map(d_in) = d_out`. Let M and M' be the multisets of rows in x and x' . Let $C = M \cap M'$, $A = M \setminus C$, and $B = M' \setminus C$. Since addition is commutative, all rows in C cancel, so

$$\text{function}(x) - \text{function}(x') = \sum_{a \in A} a - \sum_{b \in B} b.$$

By the definition of `SymmetricDistance`,

$$|A| + |B| = d_{\text{Sym}}(x, x') \leq \text{d_in}.$$

Now consider two cases.

Case 1: Known size. If `input_domain.descriptor.size` is known, then x and x' have the same number of rows, so $|A| = |B|$. Let $m = |A| = |B|$. Then $2m \leq \text{d_in}$, so $m \leq \lfloor \text{d_in}/2 \rfloor$. Enumerate $A = \{a_1, \dots, a_m\}$ and $B = \{b_1, \dots, b_m\}$. Since every row lies in the L^p ball of radius R centered at O ,

$$\begin{aligned} \|\text{function}(x) - \text{function}(x')\|_p &= \left\| \sum_{i=1}^m (a_i - b_i) \right\|_p \\ &\leq \sum_{i=1}^m \|a_i - b_i\|_p && \text{by triangle inequality} \\ &\leq \sum_{i=1}^m (\|a_i - O\|_p + \|b_i - O\|_p) \\ &\leq \sum_{i=1}^m 2R \\ &= 2mR \\ &\leq 2\lfloor \text{d_in}/2 \rfloor R. \end{aligned}$$

This is exactly the known-size stability map returned by `make_np_sum`.

Case 2: Unknown size. If `input_domain.descriptor.size` is unknown, then `make_np_sum` sets

$$\text{stability_map}(\text{d_in}) = \text{d_in} \cdot (R + \|O\|_p).$$

For every row v in either A or B ,

$$\|v\|_p \leq \|v - O\|_p + \|O\|_p \leq R + \|O\|_p.$$

Therefore,

$$\begin{aligned} \|\text{function}(x) - \text{function}(x')\|_p &= \left\| \sum_{a \in A} a - \sum_{b \in B} b \right\|_p \\ &\leq \sum_{a \in A} \|a\|_p + \sum_{b \in B} \|b\|_p && \text{by triangle inequality} \\ &\leq (|A| + |B|)(R + \|O\|_p) \\ &\leq \text{d_in}(R + \|O\|_p). \end{aligned}$$

This is exactly the unknown-size stability map returned by `make_np_sum`.

In both cases, `function(x)` and `function(x')` are `d_out`-close under `output_metric`. □